

Product Reviews & Ratings Analysis

Amazon Consumer Reviews Dataset

Reviews: 5,000 | Categories: 23 | Brand: Amazon
Avg Rating: 4.60 | Recommend Rate: 95.3%
Period: 2014-10-24 ~ 2018-09-15

Key Metrics

Metric	Value
Avg Rating	4.60
5-Star Share	69.6%
Recommend Rate	95.3%
Categories	23
Total Reviews	5,000
Avg Helpful	0.31

Key Findings

Rating Distribution

5-star: 3478 (69.6%)

4-star: 1208 (24.2%)

3-star: 197 (3.9%)

2-star: 54 (1.1%)

1-star: 63 (1.3%)

Review Depth & Helpfulness

1-star reviews are the longest (mean 300 chars) and most helpful (mean 2.63).

5-star reviews are the shortest (mean 152 chars) and least helpful (mean 0.22).

1-star reviews have 2x the length and 12x the helpfulness of 5-star reviews.

Risk Categories

eBook Readers has the highest 1-star rate at 4.2% (vs overall avg of 1.3%).

Top 5 categories by 1-star rate: eBook Readers, E-Readers & Accessories (4.2%), Tablets, Fire Tablets, Computers (3.8%), Fire Tablets, Tablets, Computers (3.4%), Computers/Tablets & Networking (2.8%), Fire Tablets, Computers/Tablets (2.7%).

Recommendation Analysis

4765 recommended (95.3%) vs 235 not recommended (4.7%).

The 235 non-recommended reviews are the most valuable signal for product improvement.

Thinking Models

1. Decomposition (Breakdown)

Evidence: Avg rating 4.6 decomposed by category, year, and recommendation status. The high average masks significant category-level differences.

Takeaway: Always decompose high-level metrics. A 4.6 average tells you little about individual product issues.

2. Segment Divergence

Evidence: Category avg ratings range from 4.41 to 5.00. Review length varies 2x between 1-star and 5-star reviews.

Takeaway: Averages flatten reality. Segment-level analysis reveals where the real problems are.

3. Proxy Inference

Evidence: Review length and helpfulness count are proxies for review quality. A short 5-star review is less informative than a detailed 1-star review.

Takeaway: Use proxy metrics (length, helpfulness) to identify high-value reviews for detailed analysis.

4. Leverage Point Identification

Evidence: Low-rated reviews (1-2 stars) are only 117 out of 5000 but contain the highest-value product improvement signals.

Takeaway: Small segments with high leverage (low volume, high value) deserve disproportionate attention.

5. Constraint vs. Preference

Evidence: A low rating is a preference signal (product did not meet expectations). Non-recommendation is closer to a constraint signal (user believes others should not buy). Different analytical responses apply.

Takeaway: Preference signals suggest product fixes; constraint signals suggest positioning or targeting fixes.